

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	8
MDMT Summary:	9
Warnings and Errors:	11
Input Echo:	12
XY Coordinate Calculations:	20
Internal Pressure Calculations:	21
Element: Btm_head	22
Element: Shell	24
Element: Shell-2	26
Element: Top_head	28
Shop Pressure Test Results:	30
Field Pressure Test Results:	32
External Pressure Calculations:	34
Element and Detail Weights:	35
Nozzle Flange MAWP:	38
Natural Frequency Calculation:	40
Forces/Moments Applied to Vessel	41
Wind Load Calculation:	42
Earthquake Load Calculation:	45
User Force/Moment Shear and Bendi	47
Wind/Earthquake Shear, Bending:	48
Wind/Earthquake Deflection:	49
Longitudinal Stress Constants:	50
Longitudinal Allowable Stresses:	51
Longitudinal Stresses due to:	52
Stress due to Combined Loads:	54
Skirt Stress Calculations:	59
Center of Gravity Calculation:	64
Basering Calculations:	65
Nozzle Calcs.: D1_2"	69
Nozzle Calcs.: A1_28"	74
Nozzle Calcs.: M1_24"	84
Nozzle Calcs.: L1_2"	89
Nozzle Calcs.: L3_2"	92
Nozzle Calcs.: L5_2"	95
Nozzle Calcs.: L7_2"	98
Nozzle Calcs.: L2_2"	101
Nozzle Calcs.: L4_2"	104
Nozzle Calcs.: L6_2"	107
Nozzle Calcs.: L8_2"	110
Nozzle Calcs.: B1_28"	113



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL24CM002KOD Ammonia comp 3rd stage7%.pvdb

Date of Analysis : Sep 18, 2026 2:03pm

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3100.000 mm ID	
Vessel Design Length, Tangent to Tangent	3850.00 mm
Distance of Bottom Tangent above Grade	2300.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	2000.00 mm
Internal Design Temperature	85 °C
Internal Design Pressure	2.500 MPa
External Design Temperature	85 °C
Maximum Allowable Working Pressure	2.51 MPa
Shop Test Pressure	3.259 MPa
Field Test Pressure	3.259 MPa
Required Minimum Design Metal Temperature	-40.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Skirt	SA-285 C	...	...	K02801	
Skirt	SA-516 70	...	...	K02700	+N+IT
Nozzle	SA-350 LF1	1	...	K03009	+IT
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Basering	SA-285 C	...	...	K02801	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.5000	N/A	No
Skirt-1	...	...	...	1.5000	N/A	No
Btm_head	2.502	0.00	2.530	1.5000	No	No
Shell	2.502	0.00	3.540	1.5000	N/A	No
Shell-2	2.500	0.00	2.510	1.5000	N/A	No
Top_head	2.500	0.00	3.240	1.5000	No	No

Liquid Level: 400.00 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.620

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-1000.0	1000.0	14.0	14.0	...	...	0.70	0.70

Skirt	0.0	1000.0	14.0	14.0	...	...	0.70	0.70
Ellipse	50.0	50.0	36.0	30.0	29.6	...	1.00	1.00
Cylinder	3050.0	3000.0	42.0	42.0	29.9	...	1.00	1.00
Cylinder	3800.0	750.0	30.0	30.0	29.9	...	1.00	1.00
Ellipse	3850.0	50.0	46.0	38.0	29.6	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
D1_2"	WRC-107/537	0.891	n/a	Passed
A1_28"	WRC-297	0.703	Shell	Passed
B1_28"	WRC-107/537	0.695	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	500	67079.1	36032.2	306795360	171449280
20	30	1500	63867.1	35821.5	241295744	135507888
30	40	2025	60655.2	35411	179009360	99877224
40	50	3550	60494.4	33634	175979392	98150408
50	60	5425	33240.9	19315.6	35319456	18693848
60	70	5825	30732.7	15397.3	11319610	5671219

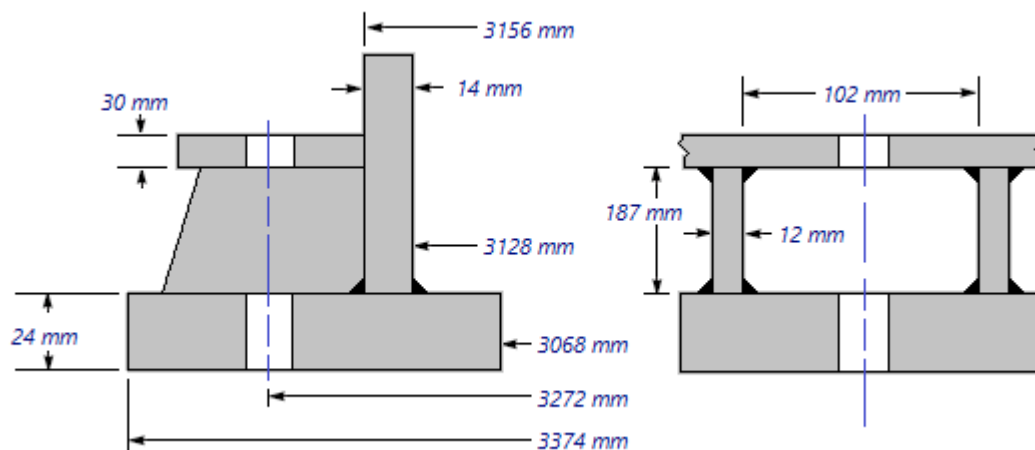
Abs Max of the all of the Stress Ratio's : 0.4947

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	24.0000 mm
Inside Diameter of Basing	3068.0000 mm
Outside Diameter of Basing	3373.9998 mm
Nominal Diameter of Bolts	30.0000 mm
Diameter of Bolt Circle	3271.9998 mm
Number of Bolts	8

Thickness of Gusset Plates	12.0000 mm
Average Width of Gusset Plates	109.0000 mm
Height of Gussets	187.0000 mm
Distance between Gussets	102.0000 mm
Thickness of Top Plate or Ring	30.0000 mm
Radial Width of the Top Plate	109.0000 mm
Number of Gussets	16

Basing Sketch for Cylindrical Skirts:





Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	128679.	N
Total Earthquake Shear on Support	97632.	N
Wind Moment on Support	829630016.	N-mm
Earthquake Moment on Support	694284032.	N-mm
Wind Load Case Test Moment on Support	276794784.	N-mm
Earthquake Load Case Test Moment on Support	4502777.	N-mm
Test Case Wind Shear on Support	42464.	N
Maximum Initial Bolt Load	77366.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	27422.3	kgm
Shop Test - Fabricated + water (full)	64264.3	kgm
Shipping - Fab. + rem. intls.+ shipping app.	34780.6	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	35700.6	kgm
Empty - Fab. + intls. + details + wghts.	35700.6	kgm
Operating - Empty + operating liquid (no ca)	37571.6	kgm
Field Test - Empty weight + test medium (full)	71622.5	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D1_2"	2.53	...	...	OK	6.30	OK	No Calc[*]
A1_28"	2.61	...	...	OK	9.83	OK	Passed
M1_24"	2.77	...	...	...		OK	Passed
L1_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
L3_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
L5_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
L7_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
L2_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
L4_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
L6_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
L8_2"	3.54	...	...	OK	6.30	OK	No Calc[*]
B1_28"	2.64	...	...	OK	9.83	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2.53 Nozzle : D12"
 Minimum MAWP Shells/Flanges : 2.51 Element : Shell-2
 Minimum MAPnc Shells/Flanges : 2.64 Element : Shell-2

Computed Vessel M.A.W.P. : 2.51 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	D1_2"	0.000	0.000	155.800
40	A1_28"	1350.000	180.000	1380.400
40	M1_24"	3000.000	0.000	1169.200
40	L1_2"	800.000	75.000	165.328
40	L3_2"	800.000	45.000	165.328
40	L5_2"	800.000	100.000	165.328
40	L7_2"	800.000	130.000	165.328
40	L2_2"	300.000	230.000	165.328
40	L4_2"	300.000	250.000	165.328
40	L6_2"	300.000	270.000	165.328
40	L8_2"	300.000	300.000	165.328
60	B1_28"	0.000	0.000	1380.400

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	230.43	600
L1_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
L3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
L5_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
L7_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
L2_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
L4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
L6_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
L8_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	242.57	300
M1_24"	24.000 in	Actual	WNF	24.000	14.000	...	...	372.26	300
A1_28"	28.000 in	Actual	WNF	28.000	12.000	...	...	383.34	300
B1_28"	28.000 in	Actual	WNF	28.000	12.000	...	...	360.51	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1_2"	SA-350 LF1	24.000	10.000	...	...	...
L1_2"	SA-350 LF1	24.000	10.000	...	...	...
L3_2"	SA-350 LF1	24.000	10.000	...	...	...
L5_2"	SA-350 LF1	24.000	10.000	...	...	...
L7_2"	SA-350 LF1	24.000	10.000	...	...	...
L2_2"	SA-350 LF1	24.000	10.000	...	...	...
L4_2"	SA-350 LF1	24.000	10.000	...	...	...
L6_2"	SA-350 LF1	24.000	10.000	...	...	...
L8_2"	SA-350 LF1	24.000	10.000	...	...	...
M1_24"	SA-350 LF1	24.000	10.000	...	...	...
A1_28"	SA-350 LF1	24.000	10.000	...	...	...
B1_28"	SA-350 LF1	24.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
D1_2"	...	0.0	200.00	0.00	Btm_head
L1_2"	800.000	75.0	200.00	0.00	Shell
L3_2"	800.000	45.0	200.00	0.00	Shell
L5_2"	800.000	100.0	200.00	0.00	Shell
L7_2"	800.000	130.0	200.00	0.00	Shell
L2_2"	300.000	230.0	200.00	0.00	Shell
L4_2"	300.000	250.0	200.00	0.00	Shell
L6_2"	300.000	270.0	200.00	0.00	Shell
L8_2"	300.000	300.0	200.00	0.00	Shell
M1_24"	3000.000	0.0	300.00	0.00	Shell
A1_28"	1350.000	180.0	300.00	0.00	Shell
B1_28"	...	0.0	300.00	0.00	Top_head

PV Elite 28 U2 Licensee: SL User
FileName : A150NRL24CM002_KOD Ammonia comp 3rd stage_7% Page 7 of 120
Nozzle Schedule: Step: 40 2:03pm Sep 18, 2026



HEXAGON

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 36.0mm THK X 3100.0mm ID X 50.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 46.0mm THK X 3100.0mm ID X 50.0mm	SA-516 70
1	PLATFORM: 1100mm X 1500mm WIDE	
1	LADDER: 4500.0mm	...
1	PLATFORM: 1100mm X 5176mm WIDE	
1	LADDER: 3000.0mm	...
1	CLASS 300 GR 1.4, 24.0" BLIND FLANGE(S)	SA-350 LF1
1	BASERING: 3374.0mm OD X 3068.0 ID X 24.0mm	SA-285 C
8	BASERING BOLTS: 30.0mm DIA	SA-36
8	BASERING BOLT NUTS: 30.0mm DIA	...
1	BASERING TOP RING: 3374.0mm OD X 3156.0 ID X 30.0mm	...
16	BASERING GUSSET PLATES: 109.0mm X 109.0mm X 187.0mm X 12.0mm	...
2	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3972.5mm X 1000.0mm	SA-285 C
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 1881.7mm X 1000.0mm	SA-285 C
2	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3972.5mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 1881.7mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 42.0mm THK X 3949.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 42.0mm THK X 3922.6mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 42.0mm THK X 1866.6mm X 3000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 30.0mm THK X 3936.9mm X 750.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 30.0mm THK X 1864.9mm X 750.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
	Notes							
Btmhead	[10] !	-40	-41		0.991	30.000	1.00	No
Btmhead	[7] !	-40	-50		0.827	36.000	1.00	No
Shell	[8] !	-40	-56		0.704	42.000	1.00	Yes
Shell-2	[8] !	-40	-40		1.000	30.000	1.00	No
Tophead	[10] !	-40	-53		0.773	38.000	1.00	No
Tophead	[7] !	-40	-60		0.640	46.000	1.00	No
D12"	[1] !	-29	-104	-29	0.026	24.000	1.00	No
Nozzle Flg	[5] !	-40	-40					
A128"	[1] !	-29	-29		0.714	12.000	1.00	No
Nozzle Flg	[5] !	-29	-52					
M124"	[1] !	-29	-29		0.514	14.000	1.00	No
Nozzle Flg	[5] !	-29	-52					
L12"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
L32"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
L52"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
L72"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
L22"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
L42"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
L62"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
L82"	[1] !	-29	-104	-29	0.038	16.764	1.00	No
Nozzle Flg	[5] !	-40	-40					
B128"	[1] !	-29	-29		0.714	12.000	1.00	No
Nozzle Flg	[5] !	-29	-52					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -29 -29

Required Minimum Design Metal Temperature -40.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	2.5	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-40.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh 0	mm
Distance Upwind of Crest	Lh 0	mm
Distance from Crest to the Vessel	x 0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm
Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	1.500	
Site Class	D	
Component Elevation Ratio	z/h 0.000	
Amplification Factor	Ap 2.500	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR] 0.000	
Component strength factor	[Rpo] 0.000	
Overstrength factor	[Omega0] 0.000	
Response Acceleration Factor	[SMS] 0.326	
Response Acceleration Factor	[SM1] 0.139	
Building Importance Factor	[Ie] 1.000	
Site Latitude	0.000	deg
Site Longitude	0.000	deg
M.A.W.P + Static Head to Nozzle	Y	
Consider MAP New and Cold in Noz. Design	N	
Consider External Loads for Nozzle Des.	Y	
Use ASME VIII-1 Appendix 1-9	N	
Perform Blast Load Analysis	No	
Material Database Year	Current w/Addenda or Code Year	

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	3156 mm
Diameter of Skirt at Base	3156 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-285 C
Allowable Stress, Ambient	108 MPa
Allowable Stress, Operating	108 MPa
Allowable Stress, Hydrotest	184.5 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	191.57 MPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02801
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	3156 mm
Diameter of Skirt at Base	3156 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	242 MPa
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	3/6
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Btmhead
Distance "FROM" to "TO"	50 mm



Inside Diameter	3100	mm
Element Thickness	30	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.5	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	D12"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	129.01	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1	+IT

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	3100	mm
Element Thickness	42	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	42	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2.5	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.3200001	
Material Name	SA-516 70	+N+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	8794560	mm ²
Platform Weight	21280	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4500	mm
Unit Weight of Ladder	0.49	N/mm

Platform Length (top head platform)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	400	mm
Liquid Density	0.000619	kg/cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	A128"	
Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	28	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7117	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	M124"	
Dist. from "FROM" Node / Offset dist	2950	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	9056.9	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L12"	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	75.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L32"	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L52"	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	100.0	
Blind Flange (Y/N)	N	



Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L72"	
Dist. from "FROM" Node / Offset dist	750	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	130.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L22"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	230.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L42"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	250.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L62"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	L82"	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	300.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	102.54	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	
Element From Node	40	



Detail Type	Weight
Detail ID	Clipwt
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	5197.2 N
Offset from Element Centerline	0 mm

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	750 mm
Inside Diameter	3100 mm
Element Thickness	30 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Weight
Detail ID	Demister
Dist. from "FROM" Node / Offset dist	-0.0002896 mm
Miscellaneous Weight	3628.2 N
Offset from Element Centerline	0 mm

Element From Node	50
Detail Type	Weight
Detail ID	Liftinglug
Dist. from "FROM" Node / Offset dist	-0.0002896 mm
Miscellaneous Weight	8040.9 N
Offset from Element Centerline	1700 mm

Element#	6/6
Element From Node	60
Element To Node	70
Element Type	Elliptical
Description	Tophead
Distance "FROM" to "TO"	50 mm
Inside Diameter	3100 mm
Element Thickness	38 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	46 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2.5 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.3200001
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	Yes

Element From Node	60
Detail Type	Platform
Detail ID	TOPPF



Dist. from "FROM" Node / Offset dist	1500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	12850201	mm ²
Platform Weight	46154	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	5176	mm
Platform Height	1100	mm
Platform Clearance or End Offset	2588	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	3000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	5176	mm

Element From Node	60	
Detail Type	Nozzle	
Detail ID	B128"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	28	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7117	N
Grade of Attached Flange	GR 1.4	
Nozzle Matl	SA-350 LF1 +IT	

Element From Node	60	
Detail Type	Weight	
Detail ID	Topdavit	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	5393.3	N
Offset from Element Centerline	1700	mm

Element From Node	60	
Detail Type	For./Mom.	
Detail ID	F/M:B128"	
Dist. from "FROM" Node / Offset dist	775	mm
Force in X Direction	43558	N
Force in Y Direction	-44800	N
Force in Z Direction	43558	N
Moment about X Axis	80007400	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	80007400	N-mm
Force/Moment Combination Method	SRSS	

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	...	1000	...	1000
Skirt-1	...	...	2000	...	1000
Btm_head	...	...	2050	...	50
Shell	...	...	5050	...	3000
Shell-2	...	...	5800	...	750
Top_head	...	...	5850	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2		...	14	1.5	3156	...	1
Skirt-1		...	14	1.5	3156	...	2
Btm_head		2.5024	36	1.5	3100	138	3
Shell		2.5024	42	1.5	3100	138	4
Shell-2		2.5	30	1.5	3100	138	5
Top_head		2.5	46	1.5	3100	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2		...	No Calc	No Calc	14	No Calc	1
Skirt-1		...	No Calc	No Calc	14	No Calc	2
Btm_head		2.5	2.53	2.67	30	29.649	3
Shell		2.5	3.54	3.68	42	29.9442	4
Shell-2		2.5	2.51	2.64	30	29.9163	5
Top_head		2.5	3.24	3.37	38	29.6222	6

Minimum 2.51 2.64

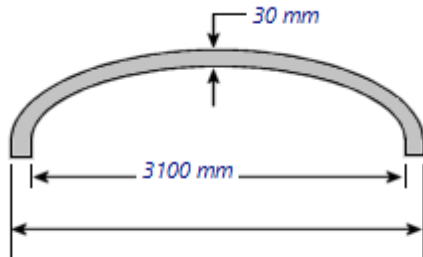
MAWP: 2.51 MPa, limited by: Shell-2.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Btmhead

Elliptical Head Design Information:					
Design Pressure	P	2.502 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3100.000 mm	Actual Stress		136.3 MPa
Spec. Min. Thk t,e, or sa		30.0000 mm	Spec. Nominal Thk.		36.000 mm
Surface Area		11404867.0 mm ²	SG of Contents		0.000
Element Volume		4277.8 ltr	Weight of Contents		0.0 N
Empty Weight		33550.6 N	Operating Weight		33550.6 N
			Comp. All. Ope. Stress		127.6 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 38.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.375 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.5 \cdot 3103 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.5)$$

$$= 28.1490 + 1.5000 = 29.6490 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.002 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 28.5) / (1 \cdot 3103 + 0.2 \cdot 28.5)$$

$$= 2.53 - 0.0024 = 2.53 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 30) / (1 \cdot 3100 + 0.2 \cdot 30)$$

$$= 2.67 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.5 \cdot (1 \cdot 3103 + 0.2 \cdot 28.5)) / (2 \cdot 1 \cdot 28.5)$$



= 136.304 MPa

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (2.5 \cdot 1552) / (138 \cdot 1 - 0.6 \cdot 2.5) + 1.5 \\ &= 29.944 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.002 MPa

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 34.5) / (1552 + 0.6 \cdot 34.5) \\ &= 3.03 - 0.0024 = 3.03 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (3103 / (2 \cdot 776))^2) / 6 \\ &= 0.998713 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 4.954 %

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 38.750 mm

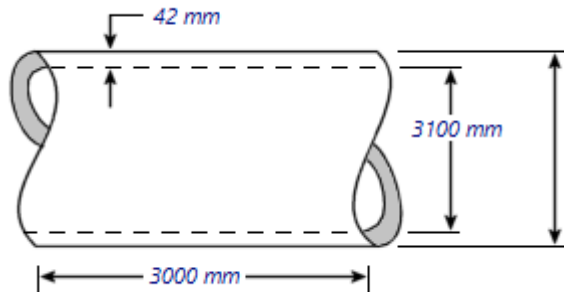
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.375 mm

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	2.502 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3100.000 mm	Actual Stress		97.4 MPa
Spec. Min. Thk t _e , or s _a		42.0000 mm	Spec. Nominal Thk.		42.000 mm
Surface Area		30008492.0 mm ²	SG of Contents		0.620
Element Volume		22647.1 ltr	Weight of Contents		18347.2 N
Empty Weight		94520.4 N	Operating Weight		112867.6 N
			Comp. All. Ope. Stress		135.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 1552) / (138 \cdot 1 - 0.6 \cdot 2.5)$$

$$= 28.4442 + 1.5000 = 29.9442 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.002 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 40.5) / (1552 + 0.6 \cdot 40.5)$$

$$= 3.55 - 0.0024 = 3.54 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 42) / (1550 + 0.6 \cdot 42)$$

$$= 3.68 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (1552 + 0.6 \cdot 40.5)) / (1 \cdot 40.5)$$

$$= 97.366 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.337 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:

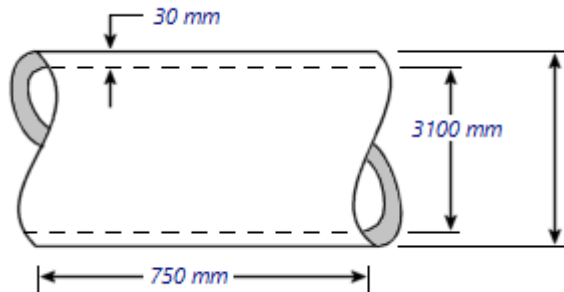
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		750.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3100.000 mm	Actual Stress		137.6 MPa
Spec. Min. Thk t _e , or s _a		30.0000 mm	Spec. Nominal Thk.		30.000 mm
Surface Area		7445575.0 mm ²	SG of Contents		0.000
Element Volume		5661.8 ltr	Weight of Contents		0.0 N
Empty Weight		16814.2 N	Operating Weight		16814.2 N
			Comp. All. Ope. Stress		127.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2.5 \cdot 1552) / (138 \cdot 1 - 0.6 \cdot 2.5)$$

$$= 28.4163 + 1.5000 = 29.9163 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 28.5) / (1552 + 0.6 \cdot 28.5)$$

$$= 2.51 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 30) / (1550 + 0.6 \cdot 30)$$

$$= 2.64 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2.5 \cdot (1552 + 0.6 \cdot 28.5)) / (1 \cdot 28.5)$$

$$= 137.596 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.958 %

Minimum Design Metal Temperature Results:



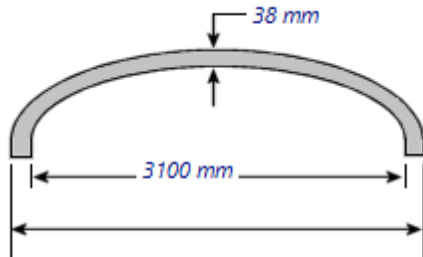
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Tophead

Elliptical Head Design Information:					
Design Pressure	P	2.500 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3100.000 mm	Actual Stress		106.4 MPa
Spec. Min. Thk t _e , or s _a		38.0000 mm	Spec. Nominal Thk.		46.000 mm
Surface Area		11545978.0 mm ²	SG of Contents		0.000
Element Volume		4277.8 ltr	Weight of Contents		0.0 N
Empty Weight		43138.5 N	Operating Weight		43138.5 N
			Comp. All. Ope. Stress		133.4 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 38.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.375 mm

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2.5 \cdot 3103 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.5)$$

$$= 28.1222 + 1.5000 = 29.6222 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 36.5) / (1 \cdot 3103 + 0.2 \cdot 36.5)$$

$$= 3.24 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 38) / (1 \cdot 3100 + 0.2 \cdot 38)$$

$$= 3.37 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2.5 \cdot (1 \cdot 3103 + 0.2 \cdot 36.5)) / (2 \cdot 1 \cdot 36.5)$$

$$= 106.380 \text{ MPa}$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)} \\ &= (2.5 \cdot 1552) / (138 \cdot 1 - 0.6 \cdot 2.5) + 1.5 \\ &= 29.916 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 44.5) / (1552 + 0.6 \cdot 44.5) \\ &= 3.89 \text{ MPa} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (3103 / (2 \cdot 776))^2) / 6 \\ &= 0.998713 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.273 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 38.750 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 19.375 mm

Material treatment options: +N+IT

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.259 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.250 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.417 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.758 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.575 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.575 MPa
App 27-4	= M.A.W.P.	2.507 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.51 • 1
 = 3.259 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	171.5	234.0	0.733	3.31
Shell	123.9	234.0	0.530	3.30
Shell-2	171.2	234.0	0.732	3.27
Top_head	133.6	234.0	0.571	3.27

Theoretical Maximum Allowable test pressure is : 4.467 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1_2"	Nozzle	118.00	118.00	1.000
A1_28"	Nozzle	118.00	118.00	1.000
M1_24"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L5_2"	Nozzle	118.00	118.00	1.000
L7_2"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
L6_2"	Nozzle	118.00	118.00	1.000
L8_2"	Nozzle	118.00	118.00	1.000
B1_28"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
D1_2"	7.01	184.50	0.038
A1_28"	96.22	184.50	0.522



M1_24"	70.00	184.50	0.379
L1_2"	6.97	184.50	0.038
L3_2"	6.97	184.50	0.038
L5_2"	6.97	184.50	0.038
L7_2"	6.97	184.50	0.038
L2_2"	6.98	184.50	0.038
L4_2"	6.98	184.50	0.038
L6_2"	6.98	184.50	0.038
L8_2"	6.98	184.50	0.038
B1_28"	95.52	184.50	0.518

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3.259 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3.250 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3.417 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	2.758 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	3.575 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3.575 MPa
App 27-4	= M.A.W.P.	2.507 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 2.51 • 1
 = 3.259 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	180.4	234.0	0.771	3.31
Shell	128.6	234.0	0.549	3.30
Shell-2	180.2	234.0	0.770	3.27
Top_head	139.0	234.0	0.594	3.27

Theoretical Maximum Allowable test pressure is : 4.243 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1_2"	Nozzle	118.00	118.00	1.000
A1_28"	Nozzle	118.00	118.00	1.000
M1_24"	Nozzle	118.00	118.00	1.000
L1_2"	Nozzle	118.00	118.00	1.000
L3_2"	Nozzle	118.00	118.00	1.000
L5_2"	Nozzle	118.00	118.00	1.000
L7_2"	Nozzle	118.00	118.00	1.000
L2_2"	Nozzle	118.00	118.00	1.000
L4_2"	Nozzle	118.00	118.00	1.000
L6_2"	Nozzle	118.00	118.00	1.000
L8_2"	Nozzle	118.00	118.00	1.000
B1_28"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
D1_2"	7.01	184.50	0.038
A1_28"	96.22	184.50	0.522



M1_24"	70.00	184.50	0.379
L1_2"	6.97	184.50	0.038
L3_2"	6.97	184.50	0.038
L5_2"	6.97	184.50	0.038
L7_2"	6.97	184.50	0.038
L2_2"	6.98	184.50	0.038
L4_2"	6.98	184.50	0.038
L6_2"	6.98	184.50	0.038
L8_2"	6.98	184.50	0.038
B1_28"	95.52	184.50	0.518

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	2000	...	...	No Calc	No Calc
20	30	2000	...	...	No Calc	No Calc
30	40	No Calc	3160	28.5	0.0012526	91.9239
40	50	4366.67	3184	40.5	0.0013837	94.6579
50	60	4366.67	3160	28.5	0.00082237	79.0599
60	70	No Calc	3176	36.5	0.0015962	98.3775

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	30	No Calc	...	0.92
40	50	42	No Calc	...	1.61
50	60	30	No Calc	...	0.95
60	70	38	No Calc	...	1.26

Minimum

0.92

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	2000	No Calc	No Calc	No Calc
20	30	2000	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	4366.67	No Calc	No Calc	No Calc
50	60	4366.67	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1649.34	...	1535.04	...	115.454
20	30	1071.01	...	956.714	...	74.9705
30	40	3421.44	4277.78	3278.88	4289.85	239.5
40	50	9639.04	22647.1	9299.22	22690.9	674.733
50	60	1714.69	5661.77	1629.73	5672.74	120.028
60	70	4399.2	4277.78	4255.75	4289.85	307.944
Total		21894	36864.43	20955	36943.38	1532

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
30	Nozl	14.0771	...	-910	...	D1_2"
40	Plat	2322.06	...	2000	-1568.39	PLAT:[1 OF 1]
40	Liqd	1871.02	...	200	...	Liquid: 20
40	Nozl	776.581	-1700	1300	...	A1_28"
40	Nozl	988.257	1700	2950	...	M1_24"
40	Nozl	11.1883	427.051	750	-1593.78	L1_2"
40	Nozl	11.1883	1166.73	750	-1166.73	L3_2"
40	Nozl	11.1883	-286.519	750	-1624.93	L5_2"
40	Nozl	11.1883	-1060.6	750	-1263.97	L7_2"
40	Nozl	11.1883	-1060.6	250	1263.97	L2_2"
40	Nozl	11.1883	-564.333	250	1550.49	L4_2"
40	Nozl	11.1883	...	250	1650	L6_2"
40	Nozl	11.1883	825	250	1428.94	L8_2"
40	Wght	530	...	...	...	Clip_wt
50	Wght	370	...	-0.00028956	...	Demister
50	Wght	820	-1700	-0.00028956	...	Lifting_lug
60	Plat	5036.18	...	1500	...	TOP_PF
60	Nozl	776.581	...	956	...	B1_28"
60	Wght	550	1700	...	...	Top_davit
60	Forc	...	...	775	...	F/M:B1_28"

Total Weight of Each Detail Type:

Platforms	7358.2
Liquid	1871.0
Nozzles	2645.0
Weights	2270.0
Sum of the Detail Weights	14144.3 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	23427.3	23427.3	23427.3	23427.3	23427.3	23427.3
Nozzles	2645.0	2645.0	2645.0	2645.0	2645.0	2645.0
Wld Weights	1350.0	1350.0	1350.0	1350.0	1350.0	1350.0
Platforms	...	...	7358.2	7358.2	7358.2	7358.2
Empty Weights	...	...	...	920.0	920.0	...
Ope Weights	...	...	...	...	...	920.0
Ope. Liquid	...	...	...	...	...	1871.0
Test Liquid	...	36841.9	...	...	...	...
Totals	27422.3	64264.3	34780.6	35700.6	35700.6	37571.6

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	27422.3 kgm
Shop Test Wt.	- Fabricated weight + water (full)	64264.3 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	34780.6 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	35700.6 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	35700.6 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	37571.6 kgm
Field Test Wt.	- Empty weight + test medium (full)	71622.5 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	20974.8 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	35110.7 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	21358.6 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	9914866	9826902
20	30	9914866	9826902
30	40	11404867	10904047
40	50	30008492	29216812
50	60	7445575	7304203
60	70	11545978	10904047

Total	80234648.0	77982912.0 mm ²
Total	80.2	78.0 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	1764.8	1764.8	1764.8	...	1764.8
20	30	1145.98	1145.98	1145.98	...	1145.98
30	40	3675.01	3675.01	7950.19	...	3675.01
40	50	15020.2	16891.2	37653.4	35700076	15020.2
50	60	3024.71	3024.71	8313.03	...	3024.71
60	70	11069.9	11069.9	14795.1	...	11069.9

Empty Support Force + the Sum of the Y forces	394879.91	N
Operating Support Force + the Sum of the Y forces	413227.06	N
Hydro Support Force + the Sum of the Y forces	747130.06	N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	35700.6	37571.6	71622.5
20	30	33935.8	35806.8	69857.7
30	40	32789.8	34660.8	68711.7
40	50	29114.8	30985.8	60761.5
50	60	14094.6	14094.6	23108.1
60	70	11069.9	11069.9	14795.1



The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	35700076	35700076	35700076
20	30	35700076	35700076	35700076
30	40	35700076	35700076	35700076
40	50	35700076	35700076	35700076
50	60	...	...	...
60	70	...	...	...

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
D1_2"	7.94	8.52	85	600	GR 1.4	6.337	5.577	7.943	5.577
A1_28"	3.97	4.26	85	300	GR 1.4	1.282	3.086	2.689	3.971
M1_24"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L1_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L3_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L5_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L7_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L2_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L4_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L6_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
L8_2"	3.97	4.26	85	300	GR 1.4	...	...	...	...
B1_28"	3.97	4.26	85	300	GR 1.4	1.282	3.086	2.689	3.971
Min Rating	3.086	4.260, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
D1_2"	77.851	0.50	524985	3200	2.507	5.577	Pass
A1_28"	762.000	0.10	102904760	44798	2.507	3.086	Pass
B1_28"	762.000	0.10	102897136	44798	2.507	3.086	Pass

Evaluating Flange on Nozzle [D12"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(524985) + 4 \cdot \text{abs}(3200)77.851 \leq \pi \cdot 77.851^3 \cdot ((7.943 - 2.507) + .500 \cdot 7.943)$$

$$= 0.93966E+07 \leq 0.13948E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle D12" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(7.943, 0.500 \cdot 7.943 + 7.943 - 9396616.0 / (\pi \cdot 77.851^3))$$

$$= \min(7.943, 5.577)$$

$$= 5.577 \text{ MPa}$$

Evaluating Flange on Nozzle [A128"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(102904760) + 4 \cdot \text{abs}(44798)762.000 \leq \pi \cdot 762.000^3 \cdot ((3.971 - 2.507) + .100 \cdot 3.971)$$

$$= 0.17831E+10 \leq 0.25878E+10 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle A128" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(3.971, 0.100 \cdot 3.971 + 3.971 - 1783075968.0 / (\pi \cdot 762.000^3))$$

$$= \min(3.971, 3.086)$$

$$= 3.086 \text{ MPa}$$

Evaluating Flange on Nozzle [B128"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(102897136) + 4 \cdot \text{abs}(44798)762.000 \leq \pi \cdot 762.000^3 \cdot ((3.971 - 2.507) + .100 \cdot 3.971)$$

$$= 0.17830E+10 \leq 0.25878E+10 \text{ Check Passes}$$



Maximum Pressure for Flange on Nozzle B128" [Pmax]:
= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
= min(3.971, 0.100 • 3.971 + 3.971 - 1782953856.0/(π • 762.000³))
= min(3.971, 3.086)
= 3.086 MPa

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 22.4801 Hz.

The Natural Frequency for the Vessel (Ope...) is 22.0517 Hz.

The Natural Frequency for the Vessel (Filled) is 16.4116 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	61600	160906928

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	61600	160906928

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
60	775.00	43558.	-44800.	43558.	0.800E+08	0.800E+08	

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.18 \cdot 0.95) / (1 + 1.7 \cdot 3.4 \cdot 0.18)))$
 $= \min(0.85, 0.9)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	22.052 Hz
Natural Frequency of Vessel (Empty)	22.480 Hz
Natural Frequency of Vessel (Test)	16.412 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.520
Structure Height to Diameter ratio	2.174
Height to top of Structure	6663.000 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Zg = 589788$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 500 + 300 = 800$ mm
 $= 2.62$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (2.62 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 589788)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf [1.59] kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.52 \cdot 44.8$$

$$= 656 \text{ lbs. [2920] N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	800.0	0.000	0.000	0.000	1.035	1.000	1.587
Skirt-1	1800.0	0.000	0.000	0.000	1.035	1.000	1.587
Btmhead	2325.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell	3850.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-2	5725.0	0.000	0.000	0.000	1.076	1.000	1.650
Tophead	6468.2	0.000	0.000	0.000	1.099	1.000	1.686

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm
 V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, $\leq 0.31416E-05$ (High); $0.31416E-05 < 0.39270E-05$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: $0/5850 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[10.4/19.2^2] \cdot 10^4 = 282$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.39270E-05$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 358571 / (5850 \cdot 3178^2)$$

$$= 0.60700E-05$$

Since Vp is $> 0.39270E-05$ no further vibration analysis is required !

Platform Load Calculations



ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	8794560.00	4350.00	1.60	14106.29	2.00
TOPPF	12850201.00	7600.00	1.71	21949.06	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Clipwt	0.00	2350.00	1.59	0.00
Demister	0.00	5350.00	1.64	0.00
Liftinglug	0.00	5350.00	1.64	0.00
Topdavit	0.00	6100.00	1.67	0.00

The Natural Frequency for the Vessel (Ope...) is 22.0517 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	800	4165.92	4165919	1.58714	3211.94
20	30	1800	4165.92	4165919	1.58714	3211.94
30	40	2325	4171.2	208560	1.58714	160.801
40	50	3850	4202.88	12608641	1.58714	27253.4
50	60	5725	4171.2	3128400	1.65044	2508.22
60	70	6468.18	4192.32	2886533	1.68585	30732.7

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	1.500
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(21.70.75)$
 = 0.201 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = $(1/\text{Natural Frequency}) = (1/22.1)$
 = 0.045

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of $(1.7 \cdot 0.2, 0.045)$ per 12.8.2
 = 0.045

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 368427 \cdot 1.5$
 = 36032.168 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 368427$
 = 16014.3 N

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 22.0517 Hz.

Earthquake Load Calculation:

	Earthquake	Earthquake	Element	Element	Element
--	------------	------------	---------	---------	---------

From	To	Height mm	Weight N	Ope Load N	Emp Load N	Vertical Load N
10	20	500	17305.6	210.703	209.429	-28785.7
20	30	1500	11237.5	410.463	407.98	-29537.9
30	40	2025	36037.2	1777.01	1766.26	-30026.4
40	50	3550	165635	14318.4	12655.4	-31592.8
50	60	5425	29660.3	3918.23	3894.54	-38792.4
60	70	5825	108551	15397.3	15304.2	-40081.6

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Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	500	61600	61600	518331936	518331936
20	30	1500	61600	61600	456706944	456706944
30	40	2025	61600	61600	395081952	395081952
40	50	3550	61600	61600	392000736	392000736
50	60	5425	61600	61600	207125680	207125680
60	70	5825	61600	61600	160906928	160906928

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	500	67079.1	36032.2	306795360	171449280
20	30	1500	63867.1	35821.5	241295744	135507888
30	40	2025	60655.2	35411	179009360	99877224
40	50	3550	60494.4	33634	175979392	98150408
50	60	5425	33240.9	19315.6	35319456	18693848
60	70	5825	30732.7	15397.3	11319610	5671219

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	61600	0.0019482	0.0076354	0.14956E-04
20	30	61600	0.016825	0.029281	0.28021E-04
30	40	61600	0.029984	0.030689	0.28305E-04
40	50	61600	0.077221	0.13051	0.37316E-04
50	60	61600	0.14471	0.15928	0.39306E-04
60	70	61600	0.16026	0.16124	0.39306E-04

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 29.250 Actual deflection : 0.161 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	67079.1	0.0011492	0.0044261	0.85128E-05
20	30	63867.1	0.0095793	0.016363	0.15039E-04
30	40	60655.2	0.01674	0.017119	0.15169E-04
40	50	60494.4	0.041557	0.068218	0.18182E-04
50	60	33240.9	0.075068	0.081966	0.18436E-04
60	70	30732.7	0.082427	0.082888	0.18436E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	1649.13	10307
20	30	1649.13	10307
30	40	1651.22	10320.1
40	50	1663.76	10398.5
50	60	1651.22	10320.1
60	70	1659.58	10372.4

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 29.250 Actual deflection : 0.083 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 29.250 Actual deflection : 0.244 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	138193	123445	108070864	96629728
20	30	138193	123445	108070864	96629728
30	40	294995	280380	228663632	217540928
40	50	414577	399961	321411744	310372896
50	60	294995	280380	228663632	217540928
60	70	374616	360001	290412512	279345824

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Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	90.7184	154.977	-101.995	-101.995
20	30	115.92	196.56	-101.995	-101.995
30	40	165.6	280.8	-127.643	-127.643
40	50	165.597	280.795	-135.5	-135.5
50	60	165.597	280.795	-127.643	-127.643
60	70	165.597	280.795	-133.356	-133.356

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	67.5482	...	88.0686
40	50	47.3858	...	61.7811
50	60	67.5482	...	88.0686
60	70	52.6336	...	68.623

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-2.83616	-2.98479	-5.6899	0.36933	0.36933
20	30	-2.69596	-2.84459	-5.5497	0.36933	0.36933
30	40	-1.14689	-1.14689	-1.14689	0.16405	0.16405
40	50	-0.71388	-0.71388	-0.71388	0.11499	0.11499
50	60	-0.49299	-0.49299	-0.49299	...	...
60	70	-0.30156	-0.30156	-0.30156	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.36933	3.17394	1.77372	1.0474	...
20	30	0.36933	2.49632	1.40189	0.82378	...
30	40	0.16405	0.82261	0.45897	0.27146	...
40	50	0.11499	0.56681	0.31613	0.18705	...
50	60	...	0.16231	0.085905	0.053561	...
60	70	...	0.040509	0.020295	0.013368	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	1.70508
20	30	...	...	...	1.35181
30	40	...	...	...	0.44496
40	50	...	...	...	0.3066
50	60	...	...	...	0.085385
60	70	...	...	...	0.020173

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
------	----	---	--



10	20	-0.36295	-0.23321
20	30	-0.36295	-0.2393
30	40	-0.1598	-0.1071
40	50	-0.11202	-0.078996
50	60	-0.1598	-0.13837
60	70	-0.12445	-0.11135

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	5.36239	5.36239	4.79469	4.79469
20	30	4.72485	4.72485	4.22464	4.22464
30	40	1.81555	1.81555	1.72723	1.72723
40	50	1.26259	1.26259	1.21923	1.21923
50	60	0.95182	0.95182	0.90552	0.90552
60	70	0.57583	0.57583	0.55389	0.55389

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.71	90.72	-12.10	102.00	0.0629	0.1187
20	4.53	115.92	-10.65	102.00	0.0391	0.1044
30	1.50	165.60	-4.11	127.64	0.0090	0.0322
40	1.12	165.60	-2.77	135.50	0.0068	0.0204
50	0.46	165.60	-1.77	127.64	0.0028	0.0138
60	0.19	165.60	-1.04	133.36	0.0011	0.0078

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.37	90.72	-10.51	102.00	0.0481	0.1030
20	3.51	115.92	-9.38	102.00	0.0303	0.0920
30	1.17	165.60	-3.68	127.64	0.0071	0.0288
40	0.89	165.60	-2.48	135.50	0.0054	0.0183
50	0.41	165.60	-1.67	127.64	0.0025	0.0131
60	0.18	165.60	-1.01	133.36	0.0011	0.0076

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.56	90.72	-12.25	102.00	0.0613	0.1201
20	4.38	115.92	-10.80	102.00	0.0378	0.1059



30	1.50	165.60	-4.11	127.64	0.0090	0.0322
40	1.12	165.60	-2.77	135.50	0.0068	0.0204
50	0.46	165.60	-1.77	127.64	0.0028	0.0138
60	0.19	165.60	-1.04	133.36	0.0011	0.0078

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.29	90.72	-10.72	102.00	0.0473	0.1051
20	3.41	115.92	-9.58	102.00	0.0294	0.0939
30	1.18	165.60	-3.69	127.64	0.0072	0.0289
40	0.90	165.60	-2.49	135.50	0.0054	0.0184
50	0.41	165.60	-1.67	127.64	0.0025	0.0131
60	0.18	165.60	-1.01	133.36	0.0011	0.0076

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-7.11	102.00		0.0697
20		196.56	-6.74	102.00		0.0661
30		280.80	-1.58	127.64		0.0124
40		280.79	-1.02	135.50		0.0075
50		280.79	-0.55	127.64		0.0043
60		280.79	-0.31	133.36		0.0024

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-6.06	102.00		0.0594
20		196.56	-5.92	102.00		0.0580
30		280.80	-1.31	127.64		0.0103
40		280.79	-0.83	135.50		0.0061
50		280.79	-0.49	127.64		0.0039
60		280.79	-0.30	133.36		0.0023

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.56	90.72	-12.25	102.00	0.0613	0.1201
20	4.38	115.92	-10.80	102.00	0.0378	0.1059
30	69.04	165.60		127.64	0.4169	
40	48.50	165.60		135.50	0.2929	
50	68.01	165.60		127.64	0.4107	
60	52.82	165.60		133.36	0.3190	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.29	90.72	-10.72	102.00	0.0473	0.1051
20	3.41	115.92	-9.58	102.00	0.0294	0.0939
30	68.73	165.60		127.64	0.4151	
40	48.29	165.60		135.50	0.2916	
50	67.95	165.60		127.64	0.4104	
60	52.82	165.60		133.36	0.3189	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.56	90.72	-12.25	102.00	0.0613	0.1201
20	4.38	115.92	-10.80	102.00	0.0378	0.1059
30	1.50	165.60	-4.11	127.64	0.0090	0.0322
40	1.12	165.60	-2.77	135.50	0.0068	0.0204
50	0.46	165.60	-1.77	127.64	0.0028	0.0138
60	0.19	165.60	-1.04	133.36	0.0011	0.0078



Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	4.29	90.72	-10.72	102.00	0.0473	0.1051
20	3.41	115.92	-9.58	102.00	0.0294	0.0939
30	1.18	165.60	-3.69	127.64	0.0072	0.0289
40	0.90	165.60	-2.49	135.50	0.0054	0.0184
50	0.41	165.60	-1.67	127.64	0.0025	0.0131
60	0.18	165.60	-1.01	133.36	0.0011	0.0076

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-7.11	102.00		0.0697
20		196.56	-6.74	102.00		0.0661
30	87.36	280.80		127.64	0.3111	
40	61.37	280.79		135.50	0.2186	
50	87.63	280.79		127.64	0.3121	
60	68.33	280.79		133.36	0.2434	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-6.06	102.00		0.0594
20		196.56	-5.92	102.00		0.0580
30	87.09	280.80		127.64	0.3101	
40	61.18	280.79		135.50	0.2179	
50	87.58	280.79		127.64	0.3119	
60	68.32	280.79		133.36	0.2433	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-3.21	102.00		0.0314
20		115.92	-3.07	102.00		0.0301
30	66.57	165.60		127.64	0.4020	
40	46.79	165.60		135.50	0.2825	
50	67.06	165.60		127.64	0.4049	
60	52.33	165.60		133.36	0.3160	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-2.67	102.00		0.0261
20		115.92	-2.54	102.00		0.0249
30	66.46	165.60		127.64	0.4013	
40	46.70	165.60		135.50	0.2820	
50	67.08	165.60		127.64	0.4051	
60	52.34	165.60		133.36	0.3161	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-3.35	102.00		0.0329
20		115.92	-3.21	102.00		0.0315
30	66.57	165.60		127.64	0.4020	
40	46.79	165.60		135.50	0.2825	
50	67.06	165.60		127.64	0.4049	
60	52.33	165.60		133.36	0.3160	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		90.72	-3.21	102.00		0.0314
20		115.92	-3.07	102.00		0.0301
30	66.57	165.60		127.64	0.4020	
40	46.79	165.60		135.50	0.2825	
50	67.06	165.60		127.64	0.4049	
60	52.33	165.60		133.36	0.3160	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-3.35	102.00		0.0329
20		115.92	-3.21	102.00		0.0315
30		165.60	-1.31	127.64		0.0103
40		165.60	-0.83	135.50		0.0061
50		165.60	-0.49	127.64		0.0039
60		165.60	-0.30	133.36		0.0023

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.51	75.60	-8.95	85.00	0.0333	0.1053
20	2.01	96.60	-8.18	85.00	0.0208	0.0962
30	68.27	138.00		106.37	0.4947	
40	47.97	138.00		112.92	0.3476	
50	67.87	138.00		106.37	0.4918	
60	52.80	138.00		111.13	0.3826	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	2.51	75.60	-8.95	85.00	0.0333	0.1053
20	2.01	96.60	-8.18	85.00	0.0208	0.0962
30	0.73	138.00	-3.23	106.37	0.0053	0.0304
40	0.58	138.00	-2.17	112.92	0.0042	0.0192
50	0.32	138.00	-1.58	106.37	0.0023	0.0149
60	0.16	138.00	-0.99	111.13	0.0012	0.0089

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-3.35	102.00		0.0329
20		115.92	-3.21	102.00		0.0315
30	66.57	165.60		127.64	0.4020	
40	46.79	165.60		135.50	0.2825	
50	67.06	165.60		127.64	0.4049	
60	52.33	165.60		133.36	0.3160	

Absolute Maximum of the all of the Stress Ratio's 0.4947

Governing Element: Btmhead

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-3.2	0.4	5.4	2.5	+ Tens
Skirt-2	...	-3.2	-0.4	-5.4	-8.9	- Comp
Skirt-1	...	-3.1	0.4	4.7	2.0	+ Tens
Skirt-1	...	-3.1	-0.4	-4.7	-8.2	- Comp
Btm_head	67.5	-1.3	0.2	1.8	68.3	+ Tens
Btm_head	67.5	-1.3	-0.2	-1.8	...	- Comp



Shell	47.4	-0.8	0.1	1.3	48.0	+ Tens
Shell	47.4	-0.8	-0.1	-1.3	...	- Comp
Shell-2	67.5	-0.6	...	1.0	67.9	+ Tens
Shell-2	67.5	-0.6	...	-1.0	...	- Comp
Top_head	52.6	-0.4	...	0.6	52.8	+ Tens
Top_head	52.6	-0.4	...	-0.6	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

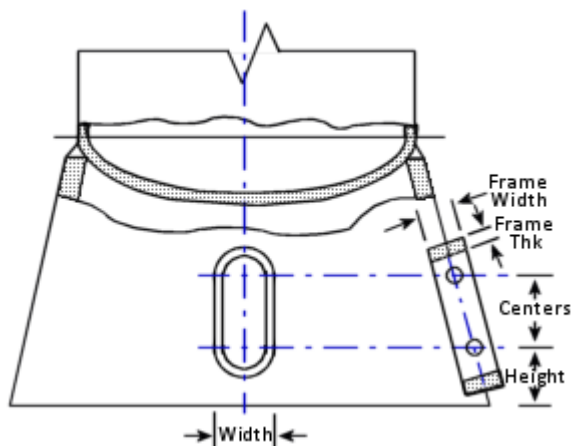
Total vessel (empty) weight	Fg	323404.125 N
Head Weight below section 2-2	DeltaFg	36037.164 N
Weight of vessel contents (liquid)	Ff	18347.186 N
Vessel Gross Weight	Fg + Ff	341751.344 N
Skirt Mean Diameter	Dz	3143.500 mm
Skirt Analysis Thickness (Corroded)	ez	12.500 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-44800.000 N
Moment at section 1-1	M1	574091328.000 N-mm
Applied Moment - Bottom of Skirt	Mb	698002688.000 N-mm
Weight at Bottom of Skirt	W	351121.438 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	3131.500 mm
Head Corroded Thickness	eb	28.500 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 776 \text{ mm} \\
 K &= D_i / (2 * h_i) = 3103 / (2 * 776) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 3103 * (0.44 * 2 + 0.02) = 2790 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 3103 * ((0.5 / 2) - 0.08) = 528 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	152478924800.000 mm ⁴
Moment of Inertia	Iyy =	152478924800.000 mm ⁴
Section Modulus	W =	97012296.000 mm ³
Section Metal Area	A =	123444.953 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{\min(I_{xx}, I_{yy}) / \text{Area}} \\
 &= \sqrt{\min(152.48E+09, 152.48E+09) / 123445} \\
 &= 1111 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 1111/1000$$

$$= 1.111$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 574091264 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -574091264/388049216 - 351121/123445$$

$$= -4.32 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +574091264/388049216 - 351121/123445$$

$$= -1.37 \text{ MPa}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= -44800 - 323404 - 18347 + 4 \cdot 574091264/3143$$

$$= 343665 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= -44800 - 323404 - 18347 - 4 \cdot 574091264/3143$$

$$= -1116767 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (343665 + 36037 + 18347) / (\pi \cdot 3132 \cdot 28.5) + 2.5 \cdot 3132 / (4 \cdot 28.5)$$

$$= 70.09 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-1116767 + 36037 + 18347) / (\pi \cdot 3132 \cdot 28.5) + 2.5 \cdot 3132 / (4 \cdot 28.5)$$

$$= 64.88 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (18347 + 36037) / (\pi \cdot 3132 \cdot 28.5) + 2.5 \cdot 3132 / (4 \cdot 28.5)$$

$$= 68.87 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 343665 / (\pi \cdot 3143 \cdot 12.5)$$

$$= 2.78 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -1116767 / (\pi \cdot 3143 \cdot 12.5)$$

$$= -9.05 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$

$$= 1 - 0.5(3132 + 28.5 - 3143 + 12.5) / (528 + 28.5)$$

$$= 0.97$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.97) = 13.1 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(28.5^2 + 12.5^2 + 2 \cdot 28.5 \cdot 12.5 \cdot 0.97)}$$

$$= 20.4 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 20.38 \cdot 343664.8 = 7.0090 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot F_{zq} = 20.38 \cdot -1116767.5 = -22.7763 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{28.5}{12.5} \right)^2$$

$$= 0.334$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.33 \cdot 6 \cdot 7008977 / (\pi \cdot 3132 \cdot 28.5^2)$$

$$= 1.76 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.33 \cdot 6 \cdot -22776256 / (\pi \cdot 3132 \cdot 28.5^2)$$

$$= -5.70 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.33 \cdot 6 \cdot 7008977 / (\pi \cdot 3143 \cdot 12.5^2)$$

$$= 9.09 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.33 \cdot 6 \cdot -22776256 / (\pi \cdot 3143 \cdot 12.5^2)$$

$$= -29.54 \text{ MPa}$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 1.12/123 = 0.0091$$

$$y = 125 \cdot eb/D_b = 125 \cdot 28.5/3132 = 1.14$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.14 = 3.97$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \cdot ((y / y_A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (2.5 + 0) \cdot 3132 \cdot ((13.1/40) \cdot 3.97 - 1) / (4 \cdot 28.5)$$

$$= 20.70 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 70.1 - 1.76 + 20.7$$

$$= 89.04 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 70.1 + 1.76 - 20.7$$

$$= 51.15 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 64.9 - 5.7 + 20.7$$

= 91.29 MPa

Stress [σ_{1qotot}]:
 $= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$
 $= 64.9 + -5.7 - 20.7$
 $= 38.48 \text{ MPa}$

Stress [σ_{2pitot}]:
 $= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$
 $= 68.9 + 1.76 + 20.7$
 $= 91.33 \text{ MPa}$

Stress [σ_{2potot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 68.9 - 1.76 - 20.7$
 $= 46.41 \text{ MPa}$

Stress [σ_{2qitot}]:
 $= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$
 $= 68.9 + -5.7 + 20.7$
 $= 83.87 \text{ MPa}$

Stress [σ_{2qotot}]:
 $= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$
 $= 68.9 - -5.7 - 20.7$
 $= 53.87 \text{ MPa}$

Stress [σ_{3pitot}]:
 $= \sigma_{3pm} - \sigma_{3pb}$
 $= 2.78 - 9.09$
 $= -6.31 \text{ MPa}$

Stress [σ_{3potot}]:
 $= \sigma_{3pm} + \sigma_{3pb}$
 $= 2.78 + 9.09$
 $= 11.88 \text{ MPa}$

Stress [σ_{3qitot}]:
 $= \sigma_{3qm} - \sigma_{3qb}$
 $= -9.05 - -29.5$
 $= 20.50 \text{ MPa}$

Stress [σ_{3qotot}]:
 $= \sigma_{3qm} + \sigma_{3qb}$
 $= -9.05 + -29.5$
 $= -38.59 \text{ MPa}$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5) \cdot (\sigma_{1pm} / f)^2)$
 $= 138(3 - (1/1.5) \cdot (70.1/138)^2)$
 $= 390.27 \text{ MPa}$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5) \cdot (\sigma_{1qm} / f)^2)$
 $= 138(3 - (1/1.5) \cdot (64.9/138)^2)$
 $= 393.66 \text{ MPa}$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5) \cdot (\sigma_{2pm} / f)^2)$
 $= 138(3 - (1/1.5) \cdot (68.9/138)^2)$
 $= 391.09 \text{ MPa}$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:
 $= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$
 $= 138(3 - (1/1.5) \cdot (68.9/138)^2)$
 $= 391.09 \text{ MPa}$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (2.78/138)^2)$$

$$= 413.96 \text{ MPa}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (-9.05/138)^2)$$

$$= 413.60 \text{ MPa}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	89.04	390.27	Pass	46
Stot1po	51.15	390.27	Pass	47
Stot1qi	91.29	393.66	Pass	48
Stot1qo	38.48	393.66	Pass	49
Stot2pi	91.33	391.09	Pass	50
Stot2po	46.41	391.09	Pass	51
Stot2qi	83.87	391.09	Pass	52
Stot2qo	53.87	391.09	Pass	53
Stot3pi	-6.31	413.96	Pass	54
Stot3po	11.88	413.96	Pass	55
Stot3qi	20.50	413.60	Pass	56
Stot3qo	-38.59	413.60	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	6274.388 mm
Center of Gravity of the Liquid	2250.000 mm
Center of Gravity of the Nozzles	4927.403 mm
Center of Gravity of the Added Weights (Operating)	4531.278 mm
Center of Gravity of the Added Weights (Empty)	4531.278 mm
Center of Gravity of Bare Shell New and Cold	3603.601 mm
Center of Gravity of Bare Shell Corroded	3621.021 mm
Vessel CG in the Operating Condition	4235.808 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4311.141 mm
Vessel CG in the Test Condition	4114.876 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3155.9998	mm
Skirt Thickness	STHK	14.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.5000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-285 C	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	24.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-285 C	
Basing Ope. All. Stress	BASOPE, f4	108.00	MPa
Basing Yield Stress		207.00	MPa
Inside Diameter of Basing	DI, D4	3068.0000	mm
Outside Diameter of Basing	DOU, D5	3373.9998	mm
Nominal Diameter of Bolts	BND, d(b0)	30.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	114.00	MPa
Number of Bolts	RN, nb	8	
Diameter of Bolt Circle	DC, DBC	3271.9998	mm
Ultimate Comp. Strength of Concrete	FPC	20.7	MPa
Allowable Comp. Strength of Concrete	FC	8.3	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	12.0000	mm
Width of Gussets at Top Plate	TWDT	109.0000	mm
Width of Gussets at Base Plate	BWDT	109.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	207.0	MPa
Height of Chair	h1	241.0000	mm
Distance between Gussets	RG, b6	102.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	30.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	109.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	36.0000	mm
External Corrosion Allowance	CA	1.5000	mm
Dead Weight of Vessel	DW	394879.9	N
Operating Weight of Vessel	ROW	413227.1	N
Test Weight of Vessel	TW	702330.2	N
Earthquake Moment on Basing	EQMOM	175952080.0	N-mm
Wind Moment on Basing	WIMOM	311298080.0	N-mm
User Moment at the Base [Earthquake Case]		518331936.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		518331936.0	N-mm (UWIMOM)
Test Moment on Basing	TM	276794784.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 694284032/3272) - 397213)/8$$

$$= 56400.2 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 829630016/3272) - 394880)/8$$

$$= 77365.8 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3272) - 413227)/8$$

$$= -51653.4 \text{ N}$$

Bolt Load, Test Moment Case:

$$= ((4 \cdot M/DC) - W)/nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 276794784/3272) - 702330)/8$$

$$= -45510.9 \text{ N}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded)	Abss :	390.7817	mm ²
Area Available in all the Bolts	Abss * RN :	3126.2537	mm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.187	0.280	1.175	2.691	0.462	0.774	91784.7	486664.6	44233.0
4	0.233	0.210	1.320	2.588	0.452	0.778	96399.0	491278.9	48300.6
6	0.198	0.192	1.212	2.665	0.459	0.775	92927.7	487807.6	45217.7
8	0.189	0.188	1.184	2.684	0.461	0.775	92069.9	486949.8	44477.3
9	0.188	0.188	1.179	2.687	0.462	0.775	91927.4	486807.2	44355.1

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 91927 / (0.3 \cdot 3272 \cdot 2.69)$$

$$= 68.8 \text{ MPa, Should be } \leq 114 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot Bolt \text{ Area} / (\pi \cdot Dc)$$

$$= 8 \cdot 391 / (\pi \cdot 3272)$$

$$= 0.3 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 1.62[(2 \cdot 0.19 \cdot 3272 + 153) / (2 \cdot 0.19 \cdot 3272)]$$

$$= 1.82 \text{ MPa, Should be } \leq 8.27 \text{ MPa}$$

Values for table 10.3, l = 109, b = 642, l/b = 0.16966155

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(2986, 10031) = 10031 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 10031/137)} + 1.5$$

$$= 22.5 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 109, b = 102, b/l = 0.93577987

As b/l (0.94) is less than 1, inverting b/l = 1.07 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1) - [(0.7 - g2)P / (4\pi)]$$

$$= 44545 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 109 \cdot \sin(\pi \cdot 58/109)) / (\pi \cdot 25))) + 1] - [(0.7 - 0.13) \cdot 44545 / (4 \cdot \pi)]$$

$$= 6199.4922 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(6199) / 138) + 1.5}$$

$$= 17.9186 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 44545 / (124 \cdot 109) + 1.5$$

$$= 11 \text{ (not less than } 9.52 + 1.5 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3272 / 8 = 1285 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (44545 \cdot 58 / (1285 \cdot 211 \cdot 162))^{2/3} \cdot 1578^{1/3} + \text{Ca}$$

$$= 3.1 + 1.5 = 4.6 \text{ mm}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	22.4905	mm
Actual Basing Thickness as entered by user	24.0000	mm
Required Thickness of Chair Cap	17.9186	mm
Actual Top Ring Thickness as entered by user	30.0000	mm
Required Gusset thickness, + CA	11.0250	mm
Actual Gusset Thickness as entered by user	12.0000	mm
Required Thickness of Skirt for Local Stress	4.5994	mm
Given Thickness of Skirt	14.0000	mm
Required Gusset Height to meet local stress	2.0517	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 12.3 \cdot (14 - 1.5)$$

$$= 153 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 153 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$$

$$= 1.41 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 44545 / (176 + 2 \cdot (187 + 30))$$

$$= 73 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 44545 \cdot 58 / (176 \cdot (217) + 0.6667 \cdot (217)^2)$$

$$= 37.1 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{73^2 + 37.1^2}$$

$$= 81.9 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 81.9 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$$



= 0.76 mm

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 44545 / (2 • 109)
= 204 N/mm

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 44545 • 58 / (2 • 217 • 109)
= 54.6 N/mm

Resultant Weld Load [Wr]:
= $\sqrt{Wv^2 + Wh^2}$
= $\sqrt{204^2 + 54.6^2}$
= 212 N/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 212 / [(0.4 • 192) • 2 • 0.707]
= 1.95 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	1.9522	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: D12"

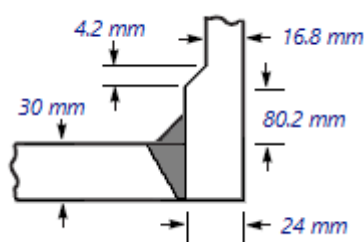
From: 30

Pressure for Reinforcement Calculations	P	2.5097	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3100.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	30.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	80.1777	mm
Height of Beveled Transition	L'	4.1777	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		24.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: D12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.51 \cdot 3103 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.51)$
 $= 25.4173 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5783 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 22.5$, less than the hub height, use (e-2)
 $= 56.2500 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	155.8000	mm
Parallel to Vessel Wall	Rn+tn+t	77.9000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	56.2500	mm

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

*Taking a UG-36(c)(3)(a) exemption for nozzle: D12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0783	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 29.7312	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 29.7312	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(29.7, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
 (see 319.4.4 and 302.3.5):**

Sustained	:	11.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	283.6 MPa	Passed
Occasional	:	1.7,	Allowable	:	156.9 MPa	Passed
Shear	:	10.0,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -29 °C
 Calculated Minimum Design Metal Temperature -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: D12"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4335 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 230.4334 mm

Input Echo, WRC107/537 Item 1, Description: D12" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	5580.000	mm
Vessel Thickness	Tv	30.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	50.800	mm
Nozzle Thickness	Tn	24.000	mm
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	2.510	MPa
Include Pressure Thrust		No	

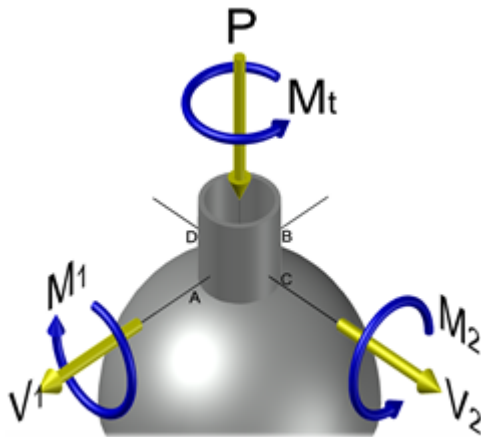
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3199.9	N
Longitudinal Shear (SUS)	(V1) V1	2828.3	N
Circumferential Shear (SUS)	(Vc) V2	2828.3	N
Circumferential Moment (SUS)	(Mc) M1	371220.7	N-mm
Longitudinal Moment (SUS)	(Ml) M2	371220.7	N-mm
Torsional Moment (SUS)	Mt	480169.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 98.8 + 2 \cdot 1.65 \cdot \sqrt{(2805.75 (30.0 - 1.5))}$$

$$= 1031.970 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-3199.9	N
Circumferential Shear	(VC) V2	2828.3	N
Longitudinal Shear	(VL) V1	2828.3	N
Circumferential Moment	(MC) M1	371220.7	N-mm
Longitudinal Moment	(ML) M2	371220.7	N-mm
Torsional Moment	MT	480169.5	N-mm

Dimensionless Param: U = 0.17 TAU = 5.00 (1.70) RHO = 1.27

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.14054	(A,B,C,D)
$M(x) / P$	SP 2	0.17723	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.43623	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.77867	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.43623	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.77867	(A,B,C,D)
$N(y) * T / P$	SP 2	0.19799	(A,B,C,D)
$M(y) / P$	SP 2	0.09045	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.03099	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.50082	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.03099	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.50082	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Rad. Memb. P	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Rad. Bend. P	4.2	-4.2	4.2	-4.2	4.2	-4.2	4.2	-4.2
Rad. Memb. MC	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Rad. Bend. MC	0.0	0.0	0.0	0.0	-7.5	7.5	7.5	-7.5
Rad. Memb. ML	-0.7	-0.7	0.7	0.7	0.0	0.0	0.0	0.0
Rad. Bend. ML	-7.5	7.5	7.5	-7.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-3.5	3.2	13.0	-10.5	-3.5	3.2	13.0	-10.5
Tang. Memb. P	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
Tang. Bend. P	2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang. Bend. MC	0.0	0.0	0.0	0.0	-4.9	4.9	4.9	-4.9
Tang. Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang. Bend. ML	-4.9	4.9	4.9	-4.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-2.0	3.4	7.8	-6.2	-2.0	3.4	7.8	-6.2
Shear VC	0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Tot. Shear	1.7	1.7	0.5	0.5	0.5	0.5	1.7	1.7
Str. Int.	4.6	5.1	13.0	10.5	3.6	3.8	13.5	11.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		122.9	122.9	122.9	122.9	122.9	122.9	122.9	122.9
Rad. Pl (SUS)		-0.2	-0.2	1.3	1.3	-0.2	-0.2	1.3	1.3
Rad. Q (SUS)		-3.4	3.4	11.7	-11.7	-3.4	3.4	11.7	-11.7
Long. Pm (SUS)		122.9	122.9	122.9	122.9	122.9	122.9	122.9	122.9
Long. Pl (SUS)		0.7	0.7	0.8	0.8	0.7	0.7	0.8	0.8
Long. Q (SUS)		-2.7	2.7	7.0	-7.0	-2.7	2.7	7.0	-7.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Pm (SUS)		122.9	122.9	122.9	122.9	122.9	122.9	122.9	122.9
Pm+Pl (SUS)		124.0	124.0	124.6	124.6	124.0	124.0	124.6	124.6
Pm+Pl+Q (Total)		122.1	128.0	136.0	116.8	121.1	126.7	136.4	117.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	122.92	138.00	Passed
Pm+Pl (SUS)	124.64	207.00	Passed
Pm+Pl+Q (TOTAL)	136.44	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: A128"

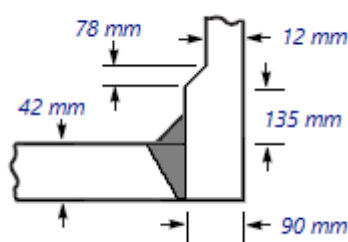
From: 40

Pressure for Reinforcement Calculations	P	2.5073	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1350.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		28.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	135.0000	mm
Height of Beveled Transition	L'	78.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: A128"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 28.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.51 \cdot 356 / (118 \cdot 1 + 0.4 \cdot 2.51)$
 $= 7.4923 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 78 / 0.5773$
 $= 135.1000 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

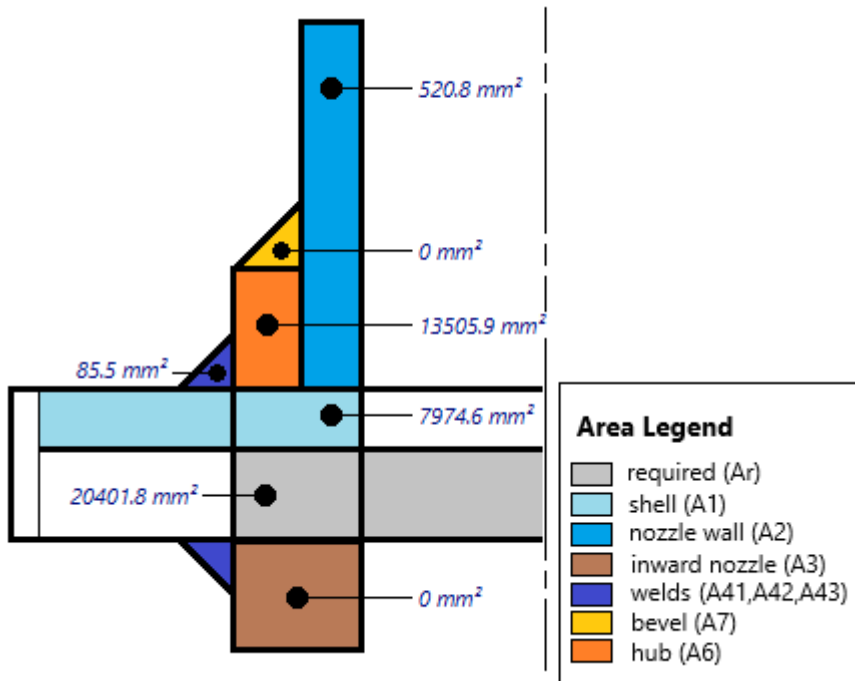
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1380.4000	mm
Parallel to Vessel Wall, opening length	d	690.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	101.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	20401.787	NA	NA
Area in Shell	A1	7974.575	NA	NA
Area in Nozzle Wall	A2	520.788	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	13505.870	NA	NA
TOTAL AREA AVAILABLE	Atot	22086.740	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (690 \cdot 28.5 \cdot 1 + 2 \cdot 88.5 \cdot 28.5 \cdot 1 \cdot (1 - 0.86))$
 $= 20401.787 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 690(1 \cdot 40.5 - 1 \cdot 28.5) - 2 \cdot 88.5$
 $(1 \cdot 40.5 - 1 \cdot 28.5) \cdot (1 - 0.86)$
 $= 7974.575 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 101)(10.5 - 7.49)0.86$
 $= 520.788 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.507 \text{ mm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$
 $= (2 \cdot \min(101, 300, 135)) \cdot (90 - 12) \cdot 0.86$
 $= 13505.870 \text{ mm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 8.9923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 30.0000 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 30.0000 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.83, \max(30, 3)]$
 $= 9.8312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(8.99, 9.83)$
 $= 9.8312 \text{ mm}$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	64.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	230.6 MPa	Passed
Occasional	:	40.6,	Allowable	:	156.9 MPa	Passed
Shear	:	16.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^{\circ}\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification $-29 \text{ }^{\circ}\text{C}$
 Flange MDMT with Temp reduction per UCS-66(i)(2) $-52 \text{ }^{\circ}\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.51/4.26 = 0.589$

Weld Size Calculations, Description: A128"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (20402 - 7975 + 2 \cdot 88.5 \cdot 0.86 \cdot (1 \cdot 40.5 - 28.5)) \cdot 138)$$

$$= 1965384.62 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (14027 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 1947278.12 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (14027 + 0 + 85.5 + (6130)) \cdot 138$$

$$= 2793073.75 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (14027 + 0 + 85.5 + 0 + (6130)) \cdot 138$$

$$= 2793073.75 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 711 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 645870. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 350) \cdot (90 - 1.5) \cdot 0.7 \cdot 118$$

$$= 8045083. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 711 \cdot (24 - 0) \cdot 0.74 \cdot 138$$

$$= 2737719. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (645870 + 8045083) = 8690953 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (645870 + 0 + 2737719 + 0) = 3383589 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (645870 + 2737719 + 0) = 3383589 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 8690953 N	, must exceed W = 1965384 N	or W1 = 1947278 N
Path 2-2 = 3383589 N	, must exceed W = 1965384 N	or W2 = 2793073 N
Path 3-3 = 3383589 N	, must exceed W = 1965384 N	or W3 = 2793073 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.61 MPa

The Drop for this Nozzle is : 41.3420 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 383.3420 mm



Input Echo, WRC297 Item 1, Description: A128"

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3100.000	mm
Shell Thickness	42.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.5000	mm
Nozzle Diameter	867.200	mm
Nozzle Thickness	90.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF1	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	2.51	MPa
Radial Load	P	-44798.03	N
Circumferential Shear	Vc	33598.50	N
Longitudinal Shear	Vl	44798.03	N
Circumferential Moment	Mc	0.627E+08	N-mm
Longitudinal Moment	Ml	0.816E+08	N-mm
Torsional Moment	Mt	0.941E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (9.636) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 867.2 + 2 \cdot 1.65 \cdot \sqrt{(1571.75(42.0 - 1.5))}$
 $= 1699.794 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3143.500	mm
Nozzle Outside Diameter	(d) =	867.200	mm
Vessel Thickness used	(T) =	40.500	mm
Nozzle Thickness used	(t) =	88.500	mm
T / t	=	0.458	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.430$$

Nr/P	=	0.064
Mr/P	=	0.123
M0/P	=	0.038
N0/P	=	0.088
MrD/Mc	=	0.275
NrDL/Mc	=	0.100



M0d/Mc = 0.084
 N0DL/Mc = 0.079
 MrD/Ml = 0.102
 NrDL/Ml = 0.059
 M0D/Ml = 0.030
 N0DL/Ml = 0.108

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	94	97	94	97
Outplane Stress Summary	86	97	119	89

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-35	35	35	-35
Inplane Pressure Stress	47	47	47	47
Inplane Stress Summary	30	60	107	-2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	86	97	120	92

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	47	47	47	47
Outplane Stress Summary	30	62	81	24



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-72	72	72	-72
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	94	97	94	97
Inplane Stress Summary	39	147	193	10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	39	147	193	25

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	13	10	13
Hoop Stress Summary	6	9	18	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	1	5	15	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 1 1 0 0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6	9	19	22

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	13	10	13

Hoop Stress Summary 9 12 16 19

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-12	12	12	-12
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary -5 13 22 -8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 0 0 1 1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14	13	22	27

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	94	97	94	97	94	97	94	97
Circ. Pl (SUS)	-4	-4	8	8	-3	-3	5	5
Circ. Q (SUS)	-4	4	16	-16	-52	52	92	-92
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	-2	-2	4	4	-1	-1	5	5
Long. Q (SUS)	-15	15	55	-55	-16	16	28	-28



Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	94.0	97.0	94.0	97.0	94.0	97.0	94.0	97.0
Pm+Pl (SUS)	90.0	93.0	102.0	105.0	91.0	94.0	99.0	102.0
Pm+Pl+Q (Total)	86.0	97.0	118.1	93.0	39.1	146.0	191.0	24.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	97.00	138.00	Passed
Pm+Pl (SUS)	105.00	207.00	Passed
Pm+Pl+Q (TOTAL)	191.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	13	10	13	10	13	10	13
Circ. Pl (SUS)	-4	-4	8	8	-1	-1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-2	-2	2	2	-1	-1	1	1
Long. Q (SUS)	-2	2	8	-8	-9	9	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	10.0	13.0	10.0	13.0	10.0	13.0	10.0	13.0
Pm+Pl (SUS)	6.0	9.0	18.0	21.0	9.0	12.0	15.0	18.0
Pm+Pl+Q (Total)	6.0	9.0	18.0	22.0	14.0	13.0	21.0	27.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	118.00	Passed
Pm+Pl (SUS)	21.00	177.00	Passed
Pm+Pl+Q (TOTAL)	27.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: M124"

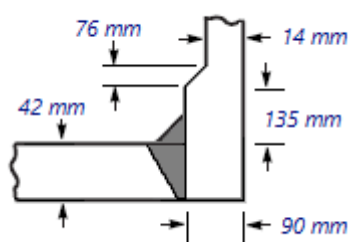
From: 40

Pressure for Reinforcement Calculations	P	2.5073	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3000.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	135.0000	mm
Height of Beveled Transition	L'	76.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M124"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 0.551 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.51 \cdot 305 / (118 \cdot 1 + 0.4 \cdot 2.51)$
 $= 6.4220 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 76 / 0.5773$
 $= 131.6359 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

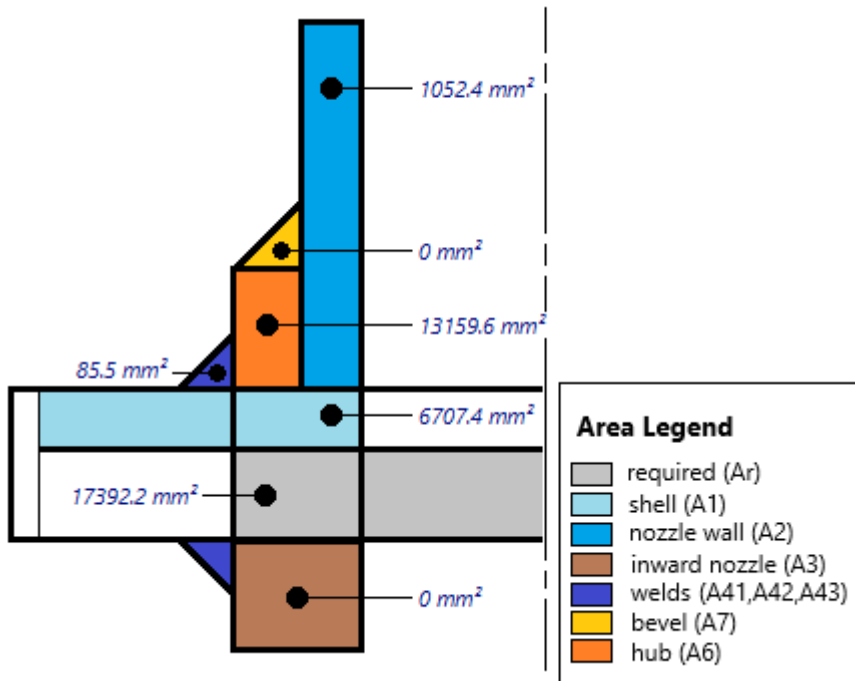
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1169.2001	mm
Parallel to Vessel Wall, opening length	d	584.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	101.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	17392.188	NA	NA
Area in Shell	A1	6707.375	NA	NA
Area in Nozzle Wall	A2	1052.422	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	13159.566	NA	NA
TOTAL AREA AVAILABLE	Atot	21004.871	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (585 \cdot 28.5 \cdot 1 + 2 \cdot 88.5 \cdot 28.5 \cdot 1 \cdot (1 - 0.86))$$

$$= 17392.188 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 585(1 \cdot 40.5 - 1 \cdot 28.5) - 2 \cdot 88.5$$

$$(1 \cdot 40.5 - 1 \cdot 28.5) \cdot (1 - 0.86)$$

$$= 6707.375 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 101)(12.5 - 6.42)0.86$$

$$= 1052.422 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$$

$$= 85.507 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(101, 300, 135)) \cdot (90 - 14) \cdot 0.86$$

$$= 13159.566 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 7.9220 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 7.9220 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification $-29 \text{ }^\circ\text{C}$

Flange MDMT with Temp reduction per UCS-66(i)(2) $-52 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 2.51/4.26 = 0.589$$

Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (17392 - 6707 + 2 \cdot 88.5 \cdot 0.86 \cdot (1 \cdot 40.5 - 28.5)) \cdot 138)$$

$$= 1724958.12 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (14212 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 1972850.88 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (14212 + 0 + 85.5 + (6130)) \cdot 138$$

$$= 2818646.75 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (14212 + 0 + 85.5 + 0 + (6130)) \cdot 138$$

$$= 2818646.75 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 553603. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 299) \cdot (90 - 1.5) \cdot 0.7 \cdot 118$$

$$= 6855600. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 610 \cdot (24 - 0) \cdot 0.74 \cdot 138$$

$$= 2346616. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (553603 + 6855600) = 7409202 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (553603 + 0 + 2346616 + 0) = 2900218 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (553603 + 2346616 + 0) = 2900218 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 7409202 N , must exceed W = 1724958 N or W1 = 1972850 N
 Path 2-2 = 2900218 N , must exceed W = 1724958 N or W2 = 2818646 N
 Path 3-3 = 2900218 N , must exceed W = 1724958 N or W3 = 2818646 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.77 MPa

The Drop for this Nozzle is : 30.2641 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 372.2641 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 584.60 mm

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Input, Nozzle Desc: L12"

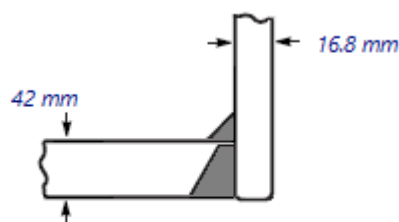
From: 40

Pressure for Reinforcement Calculations	P	2.5073	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		800.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		75.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L12"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5000 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+tr	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L12".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0777	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L12"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: L32"

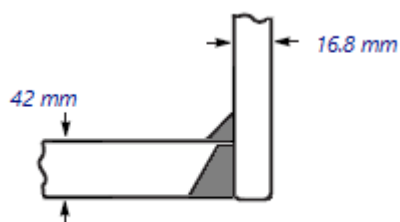
From: 40

Pressure for Reinforcement Calculations	P	2.5073	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		800.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5000 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+t	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L32".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0777	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L32"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: L52"

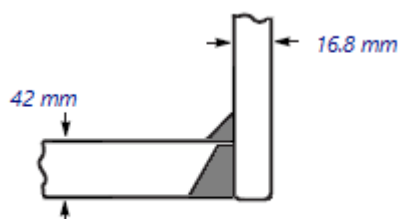
From: 40

Pressure for Reinforcement Calculations	P	2.5073	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		800.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		100.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5000$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5777$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+t	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L52".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0777	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L52"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: L72"

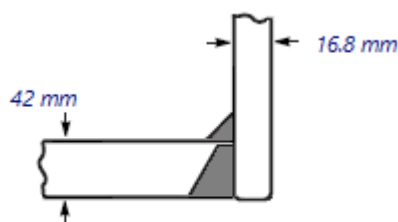
From: 40

Pressure for Reinforcement Calculations	P	2.5073	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		800.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		130.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5000$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5777$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+t	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L72".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0777	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L72"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm

PV Elite 28 U2 Licensee: SL User
FileName : A150NRL24CM002_KOD Ammonia comp 3rd stage_7% Page 100 of 120
Nozzle Calcs.: L7_2" Nozl: 31 2:03pm Sep 18, 2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: L22"

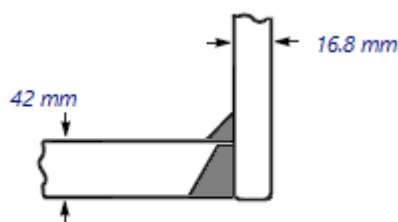
From: 40

Pressure for Reinforcement Calculations	P	2.5082	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		230.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5105 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5779 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+tr	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L22".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0779	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0105	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0105	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L22"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: L42"

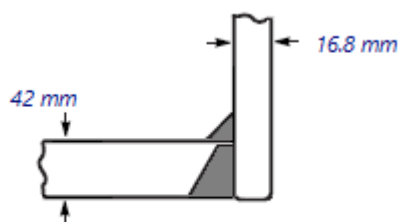
From: 40

Pressure for Reinforcement Calculations	P	2.5082	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		250.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5105 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5779 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+tr	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L42".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0779	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0105	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0105	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L42"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm

PV Elite 28 U2 Licensee: SL User
FileName : A150NRL24CM002_KOD Ammonia comp 3rd stage_7% Page 106 of 120
Nozzle Calcs.: L4_2" Nozl: 33 2:03pm Sep 18, 2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: L62"

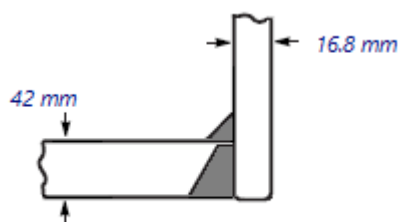
From: 40

Pressure for Reinforcement Calculations	P	2.5082	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5105 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5779 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+tr	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L62".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0779	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0105	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0105	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L62"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
---	------	---------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: L82"

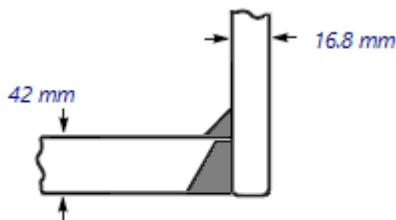
From: 40

Pressure for Reinforcement Calculations	P	2.5082	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	3100.00	mm
Shell Finished (Minimum) Thickness	t	42.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		300.00	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		300.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-350 LF1	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: L82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2.51 \cdot 1552 / (138 \cdot 1 - 0.6 \cdot 2.51)$
 $= 28.5105 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot (\exp([2.51 / (118 \cdot 1)] - 1))$
 $= 0.5779 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	165.3280	mm
Parallel to Vessel Wall	Rn+tn+t	82.6640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	38.1600	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: L82".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.0779	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.0105	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.0105	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(30, 3)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.08, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-29 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: L82"

Intermediate Calc. for nozzle/shell Welds	Tmin	15.2640	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.5736 mm



The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.5736 mm

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Input, Nozzle Desc: B128"

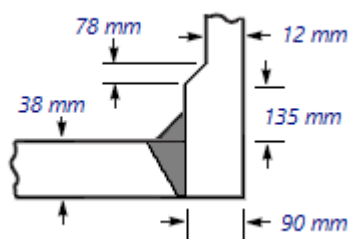
From: 60

Pressure for Reinforcement Calculations	P	2.5073	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	3100.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	38.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF1	+IT
Material UNS Number		K03009	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		28.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm
Flange Material		SA-350 LF1	+IT
Hub Height of Integral Nozzle	h	135.0000	mm
Height of Beveled Transition	L'	78.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		24.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.4	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).





ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 28.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 2.51 \cdot 3103 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 2.51)$
 $= 25.3931$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2.51 \cdot 356 / (118 \cdot 1 + 0.4 \cdot 2.51)$
 $= 7.4923$ mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 78 / 0.5773$
 $= 135.1000$ mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

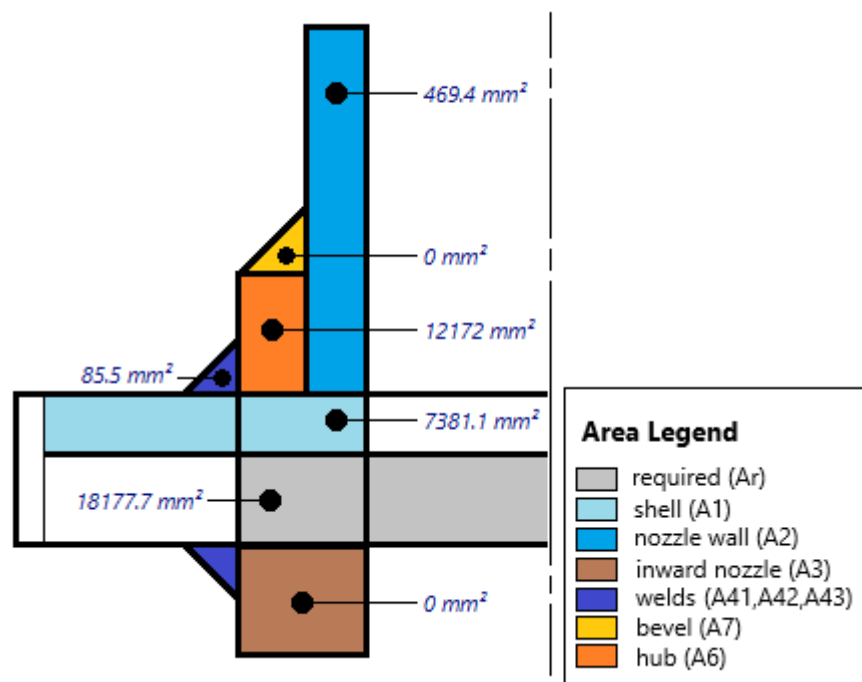
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1380.4000	mm
Parallel to Vessel Wall, opening length	d	690.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	91.2500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	18177.705	NA	NA
Area in Shell	A1	7381.066	NA	NA
Area in Nozzle Wall	A2	469.352	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	85.507	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	12171.957	NA	NA
TOTAL AREA AVAILABLE	Atot	20107.883	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (690 \cdot 25.4 \cdot 1 + 2 \cdot 88.5 \cdot 25.4 \cdot 1 \cdot (1 - 0.86))$
 $= 18177.705 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 690 (1 \cdot 36.5 - 1 \cdot 25.4) - 2 \cdot 88.5$
 $(1 \cdot 36.5 - 1 \cdot 25.4) \cdot (1 - 0.86)$
 $= 7381.066 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp) (tn - trn) fr2$
 $= (2 \cdot 91.2) (10.5 - 7.49) 0.86$
 $= 469.352 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86$
 $= 85.507 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(91.2, 300, 135)) \cdot (90 - 12) \cdot 0.86$$

$$= 12171.957 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 8.9923 \text{ mm}$
 Wall Thickness per UG16(b), $tr16b = 3.0000 \text{ mm}$
 Wall Thickness, shell/head, internal pressure $trb1 = 29.7043 \text{ mm}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 29.7043 \text{ mm}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.0000 \text{ mm}$
 Wall Thickness per table UG-45 $tb3 = 9.8312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.83, \max(29.7, 3)]$$

$$= 9.8312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(8.99, 9.83)$$

$$= 9.8312 \text{ mm}$$

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	64.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	230.6 MPa	Passed
Occasional	:	40.6,	Allowable	:	156.9 MPa	Passed
Shear	:	16.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-29 \text{ }^{\circ}\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $-29 \text{ }^{\circ}\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification $-29 \text{ }^{\circ}\text{C}$
 Flange MDMT with Temp reduction per UCS-66(i)(2) $-52 \text{ }^{\circ}\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $2.51/4.26 = 0.589$

Weld Size Calculations, Description: B128"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (18178 - 7381 + 2 \cdot 88.5 \cdot 0.86 \cdot (1 \cdot 36.5 - 25.4)) \cdot 138)$$

$$= 1721737.50 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (12641 + 85.5 - 0 \cdot 0.86) \cdot 138$$

$$= 1756119.50 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (12641 + 0 + 85.5 + (5524)) \cdot 138$$

$$= 2518380.00 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (12641 + 0 + 85.5 + 0 + (5524)) \cdot 138$$

$$= 2518380.00 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 711 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 645870. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 350) \cdot (90 - 1.5) \cdot 0.7 \cdot 118$$

$$= 8045083. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 711 \cdot (24 - 0) \cdot 0.74 \cdot 138$$

$$= 2737719. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (645870 + 8045083) = 8690953 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (645870 + 0 + 2737719 + 0) = 3383589 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (645870 + 2737719 + 0) = 3383589 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 8690953 N	, must exceed W = 1721737 N	or W1 = 1756119 N
Path 2-2 = 3383589 N	, must exceed W = 1721737 N	or W2 = 2518380 N
Path 3-3 = 3383589 N	, must exceed W = 1721737 N	or W3 = 2518380 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2.64 MPa

The Drop for this Nozzle is : 22.5070 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 360.5070 mm



Input Echo, WRC107/537 Item 1, Description: B128" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm
Vessel Diameter	Dv	5580.000	mm
Vessel Thickness	Tv	38.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.5000	mm
Nozzle Diameter	Dn	867.200	mm
Nozzle Thickness	Tn	90.000	mm
Nozzle Material		SA-350 LF1	
Nozzle UNS Number		K03009	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	2.507	MPa
Include Pressure Thrust		No	

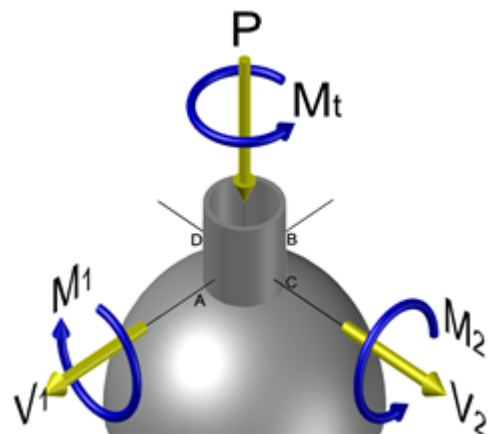
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-44798.0	N
Longitudinal Shear (SUS)	(V1) V1	39596.2	N
Circumferential Shear (SUS)	(Vc) V2	39596.2	N
Circumferential Moment (SUS)	(Mc) M1	72759256.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	72759256.0	N-mm
Torsional Moment (SUS)	Mt	94113264.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 867.2 + 2 \cdot 1.65 \cdot \sqrt{2809.75 (38.0 - 1.5)}$
 $= 1924.003 \text{ mm}$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-44798.0	N
Circumferential Shear	(VC) V2	39596.2	N
Longitudinal Shear	(VL) V1	39596.2	N
Circumferential Moment	(MC) M1	72759256.0	N-mm
Longitudinal Moment	(ML) M2	72759256.0	N-mm
Torsional Moment	MT	94113264.0	N-mm

Dimensionless Param: U = 1.35 TAU = 5.00 (4.40) RHO = 0.41

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.02947	(A,B,C,D)
M(x) / P	SP 1	0.02867	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.03037	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.04170	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.03037	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.04170	(A,B,C,D)
N(y) * T / P	SP 1	0.04397	(A,B,C,D)
M(y) / P	SP 1	0.00805	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.04142	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.01309	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.04142	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.01309	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Rad. Bend.	P	5.8	-5.8	5.8	-5.8	5.8	-5.8	5.8	-5.8
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-5.2	-5.2	5.2	5.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-42.7	42.7	42.7	-42.7
Rad. Memb.	ML	-5.2	-5.2	5.2	5.2	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-42.7	42.7	42.7	-42.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-41.1	32.7	54.6	-42.3	-41.1	32.7	54.6	-42.3
Tang. Memb.	P	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Tang. Bend.	P	1.6	-1.6	1.6	-1.6	1.6	-1.6	1.6	-1.6
Tang. Memb.	MC	0.0	0.0	0.0	0.0	-7.1	-7.1	7.1	7.1
Tang. Bend.	MC	0.0	0.0	0.0	0.0	-13.4	13.4	13.4	-13.4
Tang. Memb.	ML	-7.1	-7.1	7.1	7.1	0.0	0.0	0.0	0.0
Tang. Bend.	ML	-13.4	13.4	13.4	-13.4	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-17.4	6.2	23.6	-6.5	-17.4	6.2	23.6	-6.5
Shear	VC	0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear	VL	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear	MT	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tot. Shear		3.0	3.0	1.4	1.4	1.4	1.4	3.0	3.0
Str. Int.		41.4	33.0	54.7	42.3	41.1	32.8	54.9	42.5



WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	95.9	95.9	95.9	95.9	95.9	95.9	95.9	95.9
Rad.	Pl (SUS)	-4.2	-4.2	6.2	6.2	-4.2	-4.2	6.2	6.2
Rad.	Q (SUS)	-36.9	36.9	48.4	-48.4	-36.9	36.9	48.4	-48.4
Long.	Pm (SUS)	95.9	95.9	95.9	95.9	95.9	95.9	95.9	95.9
Long.	Pl (SUS)	-5.6	-5.6	8.5	8.5	-5.6	-5.6	8.5	8.5
Long.	Q (SUS)	-11.8	11.8	15.0	-15.0	-11.8	11.8	15.0	-15.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear	Q (SUS)	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
	Pm (SUS)	95.9	95.9	95.9	95.9	95.9	95.9	95.9	95.9
	Pm+Pl (SUS)	92.1	92.1	104.7	104.7	92.1	92.1	104.7	104.7
	Pm+Pl+Q (Total)	78.9	128.9	150.6	89.5	78.6	128.6	150.8	89.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.89	138.00	Passed
Pm+Pl (SUS)	104.67	207.00	Passed
Pm+Pl+Q (TOTAL)	150.78	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: